Vinay Kumar Singh
Senior Embedded/Network Software Engineer

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Professional Summary:

Senior Embedded Software Engineer with over 7+ years of deep expertise in Linux kernel development, custom device driver implementation, and C/C++ systems programming for ARM-based SoCs. I specialize in Board Support Package (BSP) development, creating custom Linux distributions with the Yocto Project, and architecting advanced SoC power management solutions. Proven ability to lead full-cycle projects from low-level board bring-up to OS-level performance optimization.

Educational Summary:

- Master of Technology (MTech.) Communication System from GTU with CPI **8.73** 09/2016 - 08/2018
- Master of Science (M.Sc.) Electronics from Gujarat University with **77%** 08/2013 - 07/2015
- Bachelor of Engineering (B.E.) E&TC from AM-IETE with **75%** 08/2008 - 07/2012

Technical Competencies:

- ***Board Bring-up & Feature Bring-up:*** Expertise in the bring-up of custom carrier board with Nvidia (NX & Nano) SOM, ARM , TI, Qualcomm QC6 &QC7, Samsung Exynos v920 processors, Raspberry Pi.
- ***System Initialization & Power Management:*** Proficient in early boot processes, power management, and diagnostics for multi-core clusters on Linux systems.
- ***Network Security & Protocols:*** Experience with IPsec, DPDK, Strongswan, Docker, and gRPC protocol, including unit testing with the Google Test Framework.
- ***Embedded Linux Development:*** Designed and developed custom Linux-based embedded systems using Yocto build systems, Systemd, cgroup, Cross-compilation, Recipe Development, Serial Debugging
- ***Linux* Kernel & BSP:** Linux Kernel Development, Kernel Debugging (GDB, JTAG/Lauterbach), Device Tree (DTS/DTSI), Board Bring-up, Yocto (BitBake recipes, layers), C-Boot.
- ***Programming Languages:*** Proficient in C and C++ for embedded application development.
- ***Linux Architecture & Compilation:*** In-depth understanding of Linux Kernel building, booting procedures, and cross-compilation using tool chains.
- ***Peripheral Protocols & Drivers:*** Strong knowledge of char device drivers, I2C, CAN, PCIe, and UART protocols.
- ***System Architecture:*** Expertise in Linux system architecture, including kernel space and user space, system call interfaces, and driver-to-application interactions.
- ***Operating System Fundamentals:*** Solid understanding and hand-on experience of OS fundamentals like process and thread management, multithreading, deadlocks, and memory management and sockets.
- ***Cybersecurity:*** Penetration Testing (Nmap, Nessus, OpenVAS), Vulnerability Scanning (Fortify), Security Hardening (DAC).
- ***Hardware Configuration:*** Proficient with tools like ChipLink for PCIe switch configuration and MPLAB Connect Configurator for USB 3.1 Gen 2 HUBs.

Work Experience:

<u>HARMAN International – A Samsung Company</u>,
Senior Software Engineer: 03/2023 to present.

Project: ***TI AM62 Board Bring-up & sony IMX Camera Integration — Processor SDK / Kernel***
Technologies : C, Linux Kernel, Device Tree, V4L2, GStreamer, Yocto, I²C/CSI-2, systemd

- System Optimization: Engineered an "early-boot" systemd service to parallelize camera init, significantly reducing Time-to-Image.

- Camera / Kernel: Ported and extended the IMX219 V4L2 driver to support board defaults (1080p@30), enabled 4-lane MIPI, and tuned VTS/VBLANK to double throughput from 15→30 FPS for high-throughput modes.
- Device Tree & HW: Modified device tree (DTS/DTSI) to add pinctrl, regulator entries, CSI endpoints and link frequency; integrated serializer/deserializer path and ensured proper endpoint graph for deserializer → tidss.
- Userspace & Pipelines: Built and validated GStreamer stacks combining hardware ISP (tiovxisp), KMSSINK (tidss) and v4l2h264enc for simultaneous live preview and MP4 recording; tuned encoder (bitrate, GOP) for target performance.
- Validation: Functional & stress tests using v4l2-ctl / media-ctl / gst-launch; simulated constrained CPU via U-Boot bootargs.

Project: ***SoC Power Management, Health & Thermal monitoring of Linux/Cluster VM***

Technologies : C/C++, Shell scripting, Yocto, Kernel, systemd, cgroup, Board bring-up
Operating System's : Linux, QNX QC6/QC7

- Led the design and development of SoC power management features for Android and Linux, enabling advanced low-power states like Suspend-to-RAM (S2R).
- Engineered a system-wide health and thermal monitoring daemon for automotive ECUs, polling kernel interfaces for sensor data and enforcing thermal throttling and CPU frequency throttling policies.
- Brought up and validated S2R on Samsung Exynos Auto V920 and Qualcomm QC6 & QC7, including driver + application integration.
- Performed board bring-up on Qualcomm SA6150P, enabling S2R on QNX .
- Customized Yocto BSP and image build flow by modifying .bb/.bbclass, local.conf, and bblayers.conf to enable board-specific configurations, dependencies, and feature integrations.
- Integrated cgroup v2 support into the Linux kernel and systemd to enforce fine-grained CPU and RAM resource constraints for user-space applications.
- Designed and implemented flash wear-leveling monitoring service by creating a Linux application to track and report eMMC/UFS memory block usage, enhancing storage longevity.
- Designed an early-launcher framework to initialize critical system services before systemd, meeting stringent real-time boot performance requirements.
- Implemented SMACK, DAC/MAC in Linux applications to enforce privilege separation as part of security hardening and sandboxing.
- Diagnosed and resolved critical system-level bugs, including file descriptor leaks and IPC failures, improving platform stability during S2R validation by 100%.
- Hands-on experience in executing Android CTS and ATS test cases for validating system compliance and stability.

Project: ***Android Kernel and Board Support Package (BSP) Migration***

Technologies : Embedded C, C++, Shell Scripting, Python, Git, GDB
Operating System's : Android (AOSP), Embedded Linux

- Managed Kernel, AOSP, and BSP migration on Samsung Exynos platforms.
- Engineered custom build configurations by authoring manifest files, modifying kernel defconfig, and creating new lunch targets for AOSP.
- Executed the porting of the Android kernel and BSP from legacy to target hardware, focusing on low-level driver adaptation and kernel configuration.
- Ensure functional parity post-migration, verifying core driver interfaces, system stability, and performance benchmarks.
- Resolved critical driver probe failures by patching the kernel source, modifying Device Tree Source (DTS/DTSI) files, and debugging bootloader-kernel interactions to ensure a successful boot to the UI.

INTEL Technology,
***Network Software Engineer*:** 01/2022 to 12/2022

Project: ***IPsec offloading in Smart NIC (Google MEV-TS) :***
Technologies : embedded C, Linux, C++, python, IPDK framework, DPDK, strongswan, docker
Protocol : gRPC, ipsec

- Contributed as an integral member of the ICE (Inline Cryptographic Engine) team, focusing on IPsec offloading to hardware on Intel SmartNICs.
- Engineered IPsec tunnels and transport protocols using Strongswan, configuring cryptographic policies for hardware acceleration via the IPDK framework.
- Developed a P4Runtime client application using gRPC and Protocol Buffers to dynamically program the network data plane for IPsec processing.
- Authored comprehensive unit tests using the Google Test (gTest) framework to validate the functionality and performance of the IPsec offload engine.
- Created and managed Docker containers for the development and deployment of the network stack, ensuring a consistent and reproducible environment.

eInfochips, An arrow company,
Embedded Software Engineer: 06/2018 to 12/2021

Project: ***Custom Carrier board designed for Jetson Nano and Jetson NX processor based SOM modules :***
Technologies : C, Shell scripting
IDE : MPLAB, chiplink, vim
Role : Embedded Software Developer, Configuration Manager
Duration : Dec 2020 to Dec 2021

- Developed and maintained the BSP for custom carrier boards based on Nvidia Jetson (Nano/NX) and NXP i.MX6/8 SOMs, including kernel compilation and driver integration.
- Authored custom Linux character and I2C device drivers to facilitate communication between SoCs and peripherals like FPGAs and IO expanders.
- Configured PCIe 4th Gen switches and TWI expanders using ChipLink.
- Developed firmware and managed power for a custom carrier board with a PIC Controller.
- Configured Master – slave communication of PCIe switch and TWI expander.
- Integrated drivers for I2C, CAN, and PCIe interfaces in Linux kernel/system programming.
- Implemented a "plugin manager" architecture in the BSP to dynamically apply board-specific configurations and device tree overlays at runtime using C-Boot.
- Performed runtime DTS changes using C-Boot for board-specific configurations.

Project: ***Developed Testjig using Raspberry pi 4 model B:***
Technologies : Yocto Framework, QT, Linux
Duration : Oct 2020 to Dec 2020

- Architected and developed the firmware for a Raspberry Pi 4-based automated test jig using a custom Yocto-built Linux distribution.
- Board bring up and development of Testjig to test 10000 modules using Raspberry pi 4 model B.
- Built a Qt-based application for automated control and testing.
- Integrated drivers for I2C-based IO expanders and Ambient Light Sensors (ALS).
- Wrote a suite of automation scripts (Bash) to validate peripherals including Wi-Fi, Bluetooth, Ethernet, I2C, SPI, UART, and GPIOs.
- Cross-compiled development utilities (e.g., adb, uuu) for the ARM platform using the Yocto toolchain to facilitate on-target debugging and flashing.

Project: ***Penetration testing of I.MX8M mini EVM based smart home security system:***

Technologies/OS : Penetration testing, Linux, Kali Linux
Duration : Aug 2020 to Oct 2020

- Conducted USB forensics, network, and web application penetration testing.
- Ensured product compliance with the CCDS security standard up to the 3-star level by identifying and mitigating vulnerabilities.

Project: ***I.MX6 ARM Cortex-A9-based Medical Tablet for CPA patient:***

Technologies/OS : Yocto Framework, Linux, Kali Linux
Duration : May 2020 to Aug 2020

- Performed initial board bring-up and validation for the phase-1 prototype of a medical-grade tablet.
- Authored key FDA-compliance documentation, including Hazard Analysis (HA), FMEA, HRS, SRS, and various test reports.
- Conducted penetration testing on network interfaces (Wi-Fi, Bluetooth) and applications, preparing detailed executive reports on findings.

Project: ***IMX6 Series ARCH: ARM platform, Processor Cortex-A9 core:***

Technologies : C/C++, Yocto Framework
Operating System's : Linux (Ubuntu 14.04 and Fedora)
Duration : Jun 2018 to Nov 2019

- Developed firmware using the Yocto Build System.
- Created C/C++ applications for peripherals NIC, Wi-Fi, Printers, VFD, Barcode scanners, and Smartcard readers.
- Prepare test plan and unit test cases for development and production phase for quality deliverables.
- Conducted manual and automated testing for development and production products.
- Designed web pages using JavaScript, HTML, and CSS.

einfochips, An arrow company, 06/2017 to 06/2018
Project Trainee: Worked as an Intern on live/industry project that presented as MTech dissertation.

L.D. College of Engineering, Gujarat Technological University, 06/2017 to 03/2020
Visiting faculty: Worked as part time faculty and taught Professional students.

Gujarat Arts & Science College, Gujarat University, 06/2015 to 08/2017
Visiting faculty: Worked as part time faculty and took expert lecture and labs.

Skills:

Linux/Fedora	● ● ● ● ●	C	● ● ● ● ○	Yocto Build system	● ● ● ● ○
Python	● ○ ○ ○ ○	C++	● ● ● ○ ○	Shell scripting	● ● ● ○ ○

Paper Publications:

- Published and presented a research paper titled "HID Implementation Using CYPRESS FX-3" in **IEEE** conference.
- Published research paper titled "Selection, Design and Implementation of Oversized Waveguide for Michelson Interferometer Diagnostic for SST-1 Tokamak" in **IEEE** conference.